AME 415 Group 7: Presentation 3, Critical Design Review

Abby R., Jake S., Kondwani P.

AME415 Project Grade Sheet			Reference Grading **Phase3**						
1	**Clearly Display the Design Parameters Chosen**								
	φ	ψ	Nr stators	Nr rotors	BL				
	40	40	25	25	20				
2	**Overal accuracy of 1-D calculations**								
	2.1	**Efficiency TT**	**Power**						
		150	150						
3	**Detailed accuracy of Velocity Triangles**								
	3.1	Mean	Ca	α2	α2'	α3'			
			100	100	100	100			
	3.2	Tip	Ca	α2	α2'	α3'			
			100	100	100	100			
	3.2	Hub	Ca	α2	α2'	α3'			
			100	100	100	100			
4	**Sizing of main Turbine Elements**								
	4.1	Rmean	L2	L3	o2	o3	Stage sketch		
		100	100	100	100	100	100		
5	**Airfoil Design**								
	4.1	Mean	Stator	Rotor					
			100	100					
	4.2	Tip	Stator	Rotor					
			100	100					
	4.3	Hub	Stator	Rotor					
			100	100					
6	**Airfoil Analysis CFD**		**Scene/Plot**	**Scene/Plot**	**XY Plot**	**List/Table**	**List/Table**	**List/Table**	**List/Table**
	5.1	Mean	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow
			50	50	50	50	50	50	50
	5.2	Tip	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow
			50	50	50	50	50	50	50
	5.3	Hub	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow
			50	50	50	50	50	50	50
7	**Correlate CFD analysis with 1-D analysis**								
	6.1	CFD Torque HP		200					
	6.2	CFD Euler Velocity HP		200					
8	**Accurate Transmittal of Airfoil Coordinates for 3D CFD Virtual Testing**								
	8.1	Stator Coordinates		Stator_Hub.csv	Stator_Mean.c	Stator_Tip.csv			
				50	50	50			
	8.2	Rotor Coordinates		Rotor_Hub.csv	Rotor_Mean.cs	Rotor_Tip.csv			
				50	50	50			
9	**3D CFD Virtual Test**								
	9.1	3D CFD virtual test compares team's declared performance to the 3D simulation					1000		
		using the coordinates provide for stator and rotor. 1000 are awarded to each team from start							
		Then, if 3D CFD power differs from Project Team power value by more than 5 percent, the power differences (lower ot higher)							
		is calculated as a penalty equal to the 1/5 of the HP difference value above 5% err					Similar for massflow difference		
		between 3D CFD and Project Team stated value. The two penalties are cumulative and are subtracted from the 5000 p							
TOTAL	5600								

$$\text{Power Penalty} = \frac{|\text{TP} - \text{VTP}| - 0.05 * \text{VTP}}{8}$$

$$\text{Mflow Penalty} = \frac{|\text{mTP} - \text{mVTP}| - 0.05 * \text{mVTP}}{5}$$

Where:

- TP – team's declared turbine CFD power
- VTP – virtual test power 3D CFD, using team's coordinates
- mTP – team's declared massflow, average stator/rotor
- mVTP – virtual test 3D CFD massflow

	G7
Po1 (Pa)	7584232.7
To1 (K)	400.0
P3(Pa)	6391439.0
Co (m/s)	742.2
CFD TEST 3D C3 (m/s)	208.0
work isentropic (J/Kg)	253808.2
Nr Vanes	240
Nr Blades	230
CFD TEST 3D Mdot/Vane	0.1873
CFD TEST 3D Mdot/Blade	0.1954
Mdot1 all stators	44.9
Mdot2 all rotors	44.9
Mdot average	**44.94**
HP isen (watt)	11405620.4
Torque/Blade (Nm)	30.4500
HP torque (watt)	11001072.1
CFD TEST HP 3D torque (hp)	**14752.7**
CFD TEST 3D TT Efficiency	**0.9645**
CFD TEST 3D TS Efficiency	**0.8888**
Team Reported CFD 2D HP Torque	15226.0
Team Reported CFD 2D HP Euler	15610.0
Team Reported CFD 2D HP Average	15418.0
Team Stator Mdot	48.8
Team Rotor Mdot	48.2
Team Reported CFD 2D Mdot average	48.5
HP Average error project vs. 3D CFD	4.51%
Mdot error project vs. 3D CFD	7.89%

Massflow Delta (kg/sec)	-3.5
Power Delta (hp)	-665.3
Power Penalty (points)	**0.0**
Massflow Penalty (points)	**-0.2**
6.1 CFD Force HP 3D TEST VERIF	**1000.0**

AME415 Project Grade Sheet		Team 7							
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	40	40	25	25	20				
2	**Overal accuracy of 1-D calculations**								
2.1	**Efficiency TT**	**Power**							
	150	150							
3	**Detailed accuracy of Velocity Triangles**								
3.1	Mean	Ca	α2	α2'	α3'				
	100	100	100	100	100				
3.2	Tip	Ca	α2	α2'	α3'				
	100	100	100	100					
3.2	Hub	Ca	α2	α2'	α3'				
	100	100	100	100					
4	**Sizing of main Turbine Elements**								
4.1	Rmean	L2	L3	o2	o3	Stage sketch			
	100	100	100	100	100	100			
5	**Airfoil Design**								
4.1	Mean	Stator	Rotor						
		100	100						
4.2	Tip	Stator	Rotor						
		100	100						
4.3	Hub	Stator	Rotor						
		100	100						
6	**Airfoil Analysis CFD**	**Scene/Plot**	**Scene/Plot**	**XY Plot**	**List/Table**	**List/Table**	**List/Table**	**List/Table**	
5.1	Mean	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow	
		50	50	50	50	50	50	50	
5.2	Tip	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow	
		50	50	50	50	50	50	50	
5.3	Hub	Sta P/M/E/V	Rot P/M/E/V	Rot/Sta Loadin	Sta Vel/Angles	Rot Vel/Angles	Rtor Force	Sta/Rot Mflow	
		50	50	50	50	50	50	50	
7	**Correlate CFD analysis with 1-D analysis**								
6.1	CFD Torque HP		200						
6.2	CFD Euler Velocity HP		200						
8	**Accurate Transmittal of Airfoil Coordinates for 3D CFD Virtual Testing**								
8.1	Stator Coordinates		Stator_Hub.csv	Stator_Mean.c	Stator_Tip.csv				
			50	50	50				
8.2	Rotor Coordinates		Rotor_Hub.csv	Rotor_Mean.cs	Rotor_Tip.csv				
			50	50	50				
9	**3D CFD Virtual Test**								
9.1	3D CFD virtual test compares team's declared performance to the 3D simulation					1000			
	using the coordinates provide for stator and rotor. 1000 are awarded to each team from start								
	Then, if 3D CFD power differs from Project Team power value by more than 5 percent, the power differences (lower ot higher)								
	is calculated as a penalty equal to the 1/5 of the HP difference value above 5% err					Similar for massflow difference			
	between 3D CFD and Project Team stated value. The two penalties are cumulative and are subtracted from the 5000 p								
TOTAL	5600								
TOTAL Possible	5600								
Grade	**100.0%**								

1 - Design Requirements and Operating Conditions

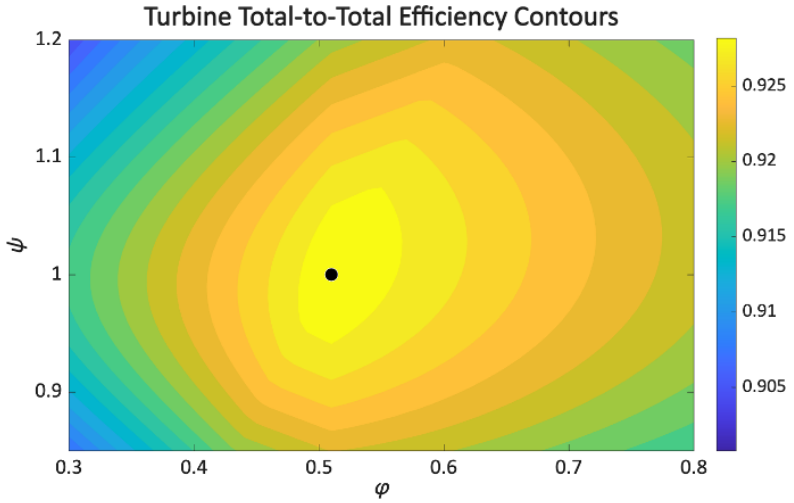
Inlet Total Pressure, *P01*	7.58423270	MPa
Inlet Total Temperature, *T01*	400	K
Available Inlet Mass Flow, *mdot*	47.6	Kg/sec
Nozzle Inlet Angle, *nozIn*	90.0	deg
Exit Static Pressure, *P3*	6.39143974	MPa
Shaft Speed, *N*	15000	rpm

Properties of working fluid, gaseous hydrogen (GH2):

a. Gamma = 1.4056
b. Rgas = 4124.4 J/Kg-K
c. Cp = 14292.3 J/Kg-K

2.7 Turbine Sizing & Selection of Phi, Psi

To select the most efficient design flow parameter ϕ and loading factor ψ, a sweep was performed on MATLAB code and the combination of values with the greatest total-to-total efficiency were selected to continue with calculations.



Values selected:

ϕ=0.5
ψ=1.05

2.8 - Element Selection and Performance Summary

Chosen Elements:

statNrVanes = 240;
rotorNrVane= 230;
BL = 0.9;

Performance:

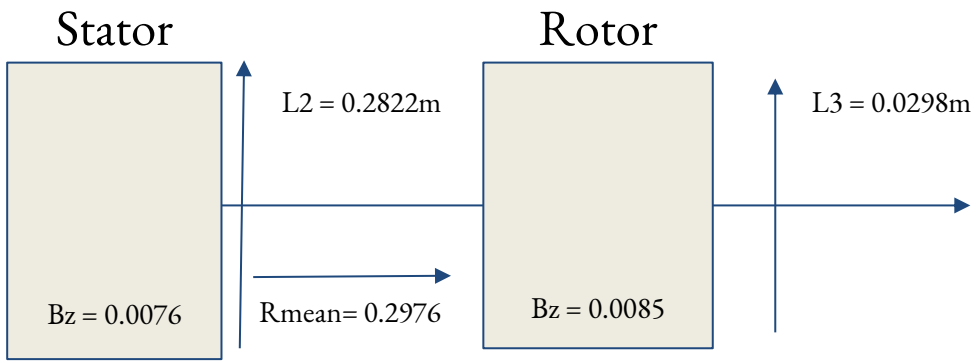
T-T Efficiency = 0.9244
Power = 14,645.9 hp
U = 467.4568 m/s
Ca = 233.7284 m/s
Co = 742.1905 m/s

2.8 - Airfoil Geometry

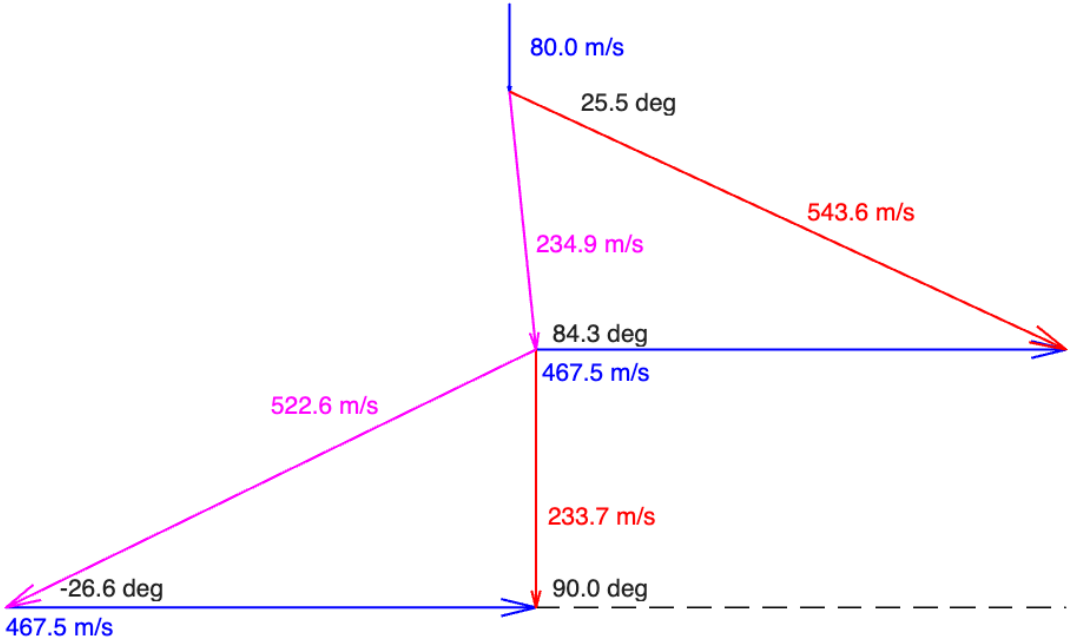
STATOR
o_2 mean = 0.00301 m
o_2 tip = 0.00328 m
o_2 hub = 0.00276 m

ROTOR
o_3 mean = 0.00327 m
o_3 tip = 0.00330 m
o_3 hub = 0.00324 m

RADII
rmean = 0.2976 m
rtip = 0.3117 m
rhub = 0.2835 m

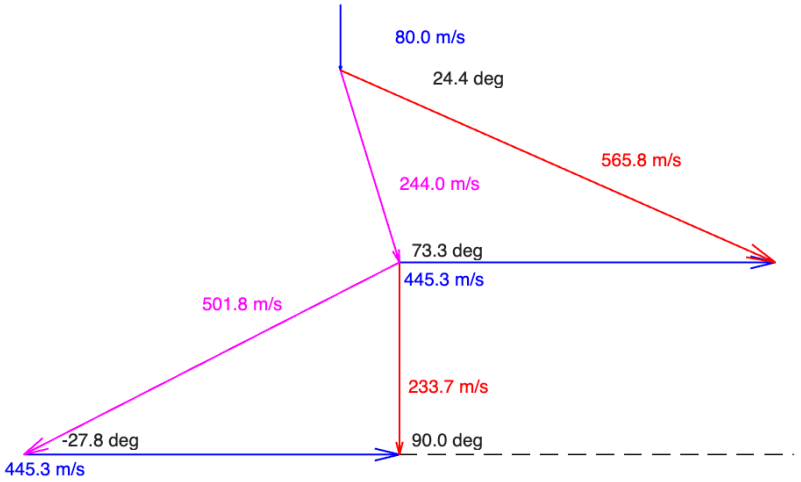


2.10 Velocity Triangles - Mean Radius

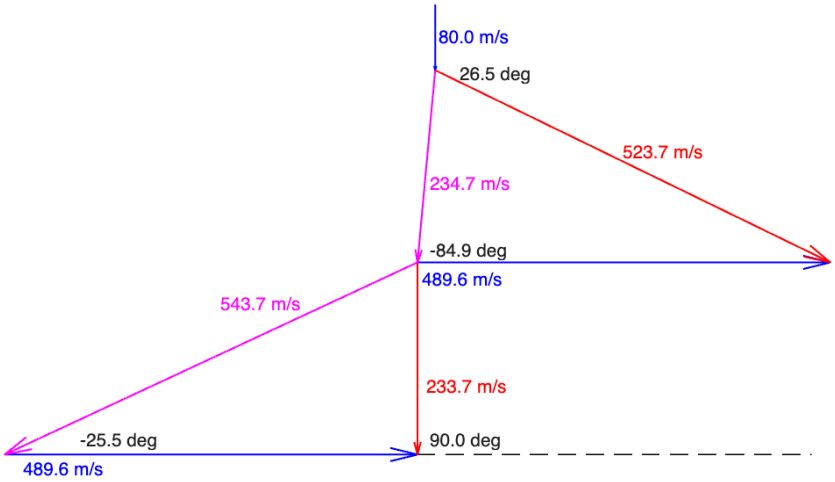


2.11 Velocity Triangles - Tip and Hub Radii

Hub Radius

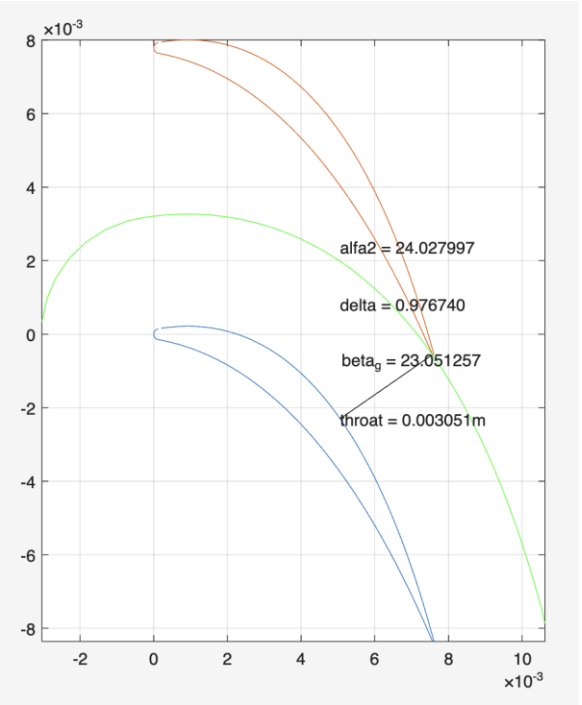


Tip Radius



2.12.1 Airfoil Geometry at Mean Radius - Stator

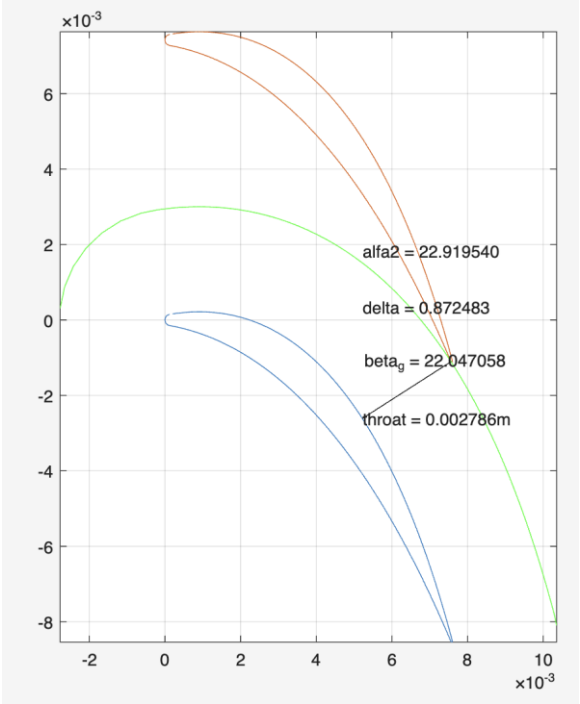
BZ	0.0076
Radius LE	0.00015
Radius TE	2e-05
Beta 1	90
Beta 2	19
Beta s	48.3
Wedge Angle 1 LE	10
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.2976
Number of blades	240
M2	0.3617



Targets:
Angle = 25.4 deg
Throat = 0.00301m

2.12.1 Airfoil Geometry at Hub Radius - Stator

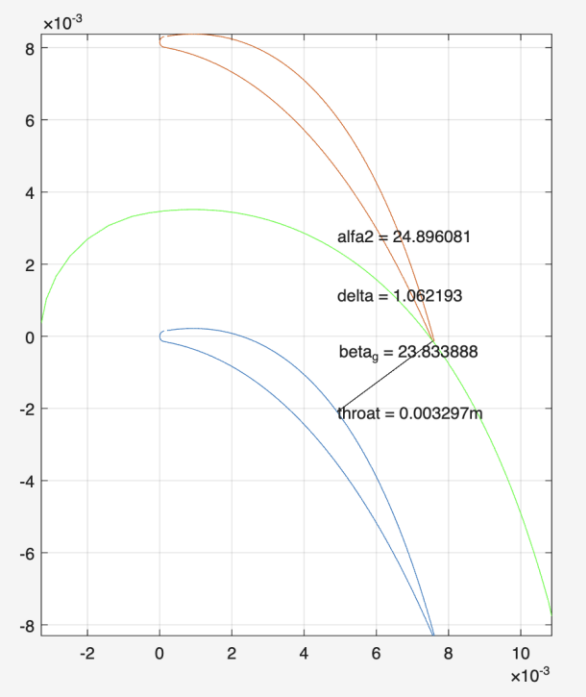
BZ	0.0076
Radius LE	0.00015
Radius TE	2e-05
Beta 1	90
Beta 2	18.8
Beta s	48.9
Wedge Angle 1 LE	10
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.2835
Number of blades	240
M2	0.3617



Targets:
Angle = 24.3 deg
Throat = 0.00276 m

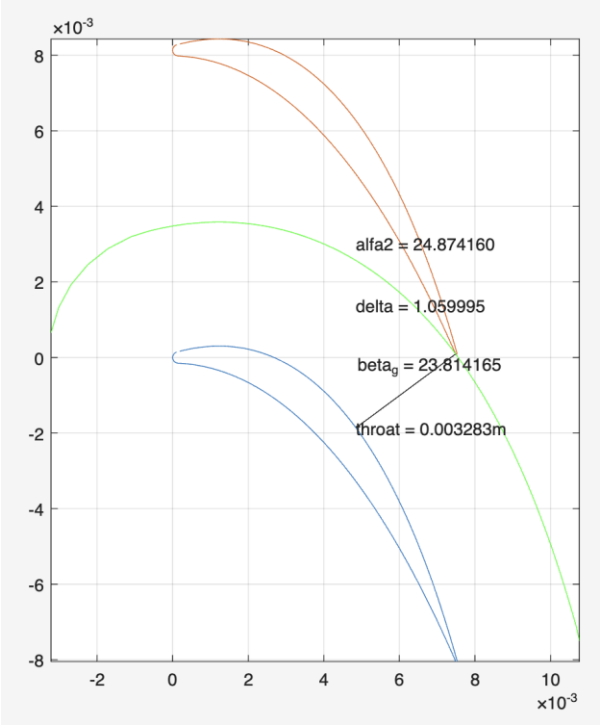
2.12.1 Airfoil Geometry at Tip Radius - Stator

BZ	0.0076
Radius LE	0.00015
Radius TE	2e-05
Beta 1	90
Beta 2	19.2
Beta s	48.1
Wedge Angle 1 LE	10
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.3117
Number of blades	240
M2	0.3617



Targets:
Angle = 26.5 deg
Throat = 0.00328 m

2.12.2 Airfoil Geometry at Mean Radius - Rotor

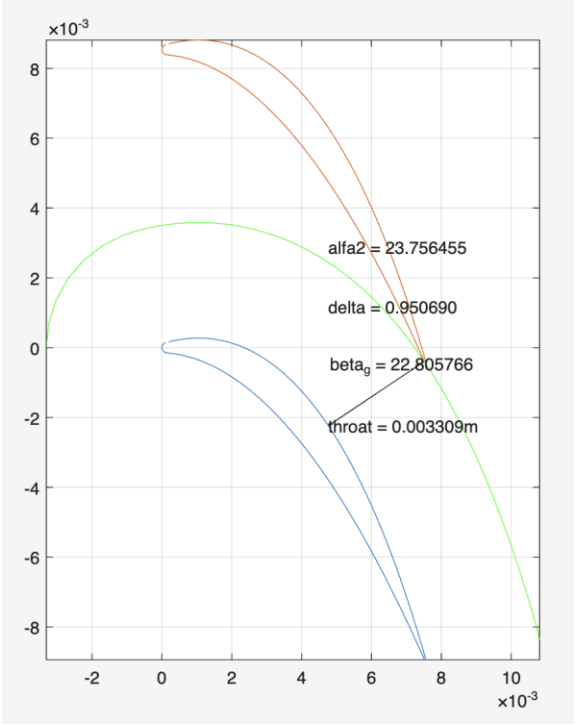


BZ	0.007525
Radius LE	0.00015
Radius TE	2e-05
Beta 1	84.4
Beta 2	19.5
Beta s	47.5
Wedge Angle 1 LE	8
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.2976
Number of blades	230
M2	0.3512

Targets:
Angle = 26.5 deg
Throat = 0.00327m

2.12.2 Airfoil Geometry at Tip Radius - Rotor

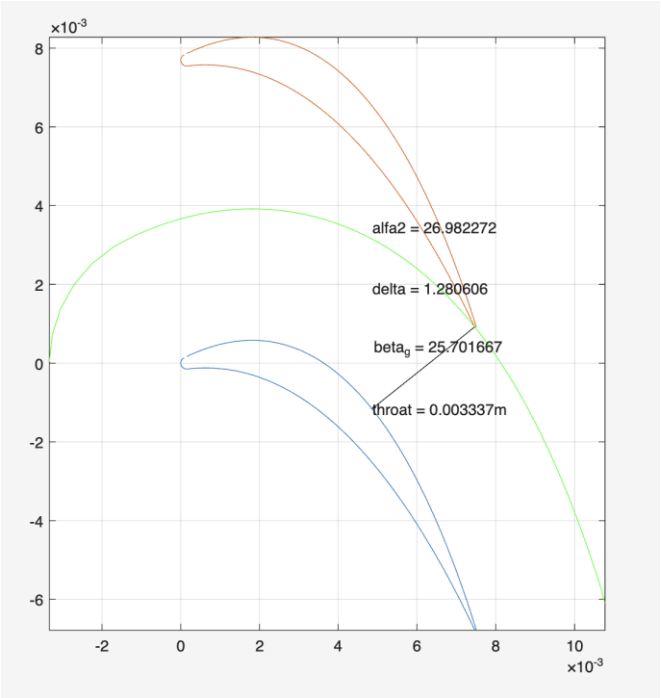
BZ	0.00755
Radius LE	0.00015
Radius TE	2e-05
Beta 1	84.4
Beta 2	19.8
Beta s	50.4
Wedge Angle 1 LE	9.8
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.3125
Number of blades	230
M2	0.3512



Targets:
Angle = 25.4 deg
Throat = 0.00330 m

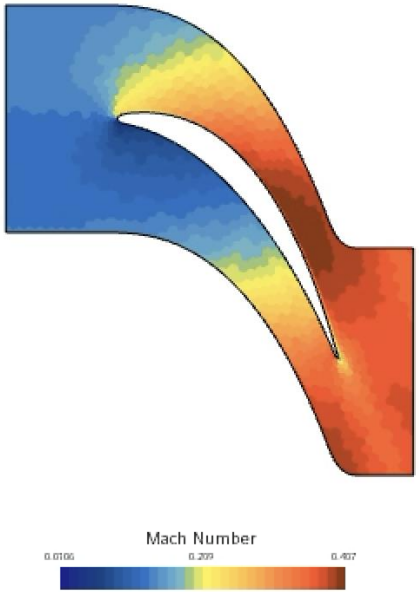
2.12.2 Airfoil Geometry at Hub Radius - Rotor

BZ	0.0075
Radius LE	0.00015
Radius TE	2e-05
Beta 1	73.3
Beta 2	21.3
Beta s	42.7
Wedge Angle 1 LE	10
Wedge angle 2 LE	10
Wedge angle 1 TE	4
Wedge angle 2 TE	4
Radius	0.2827
Number of blades	230
M2	0.3512



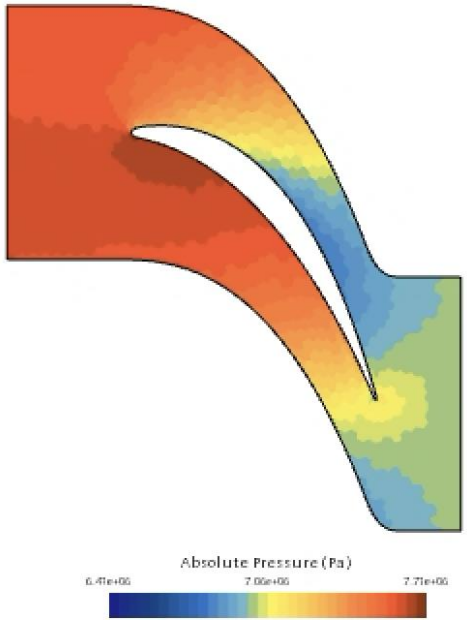
Targets:
Angle = 27.7 deg
Throat = 0.00324 m

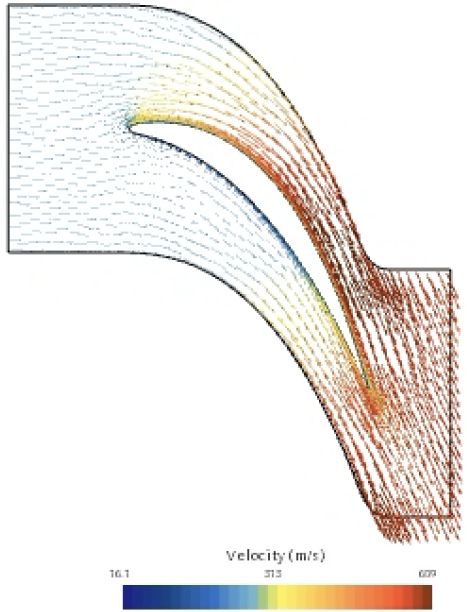
Stator Mach and Entropy Plots at Mean Radius



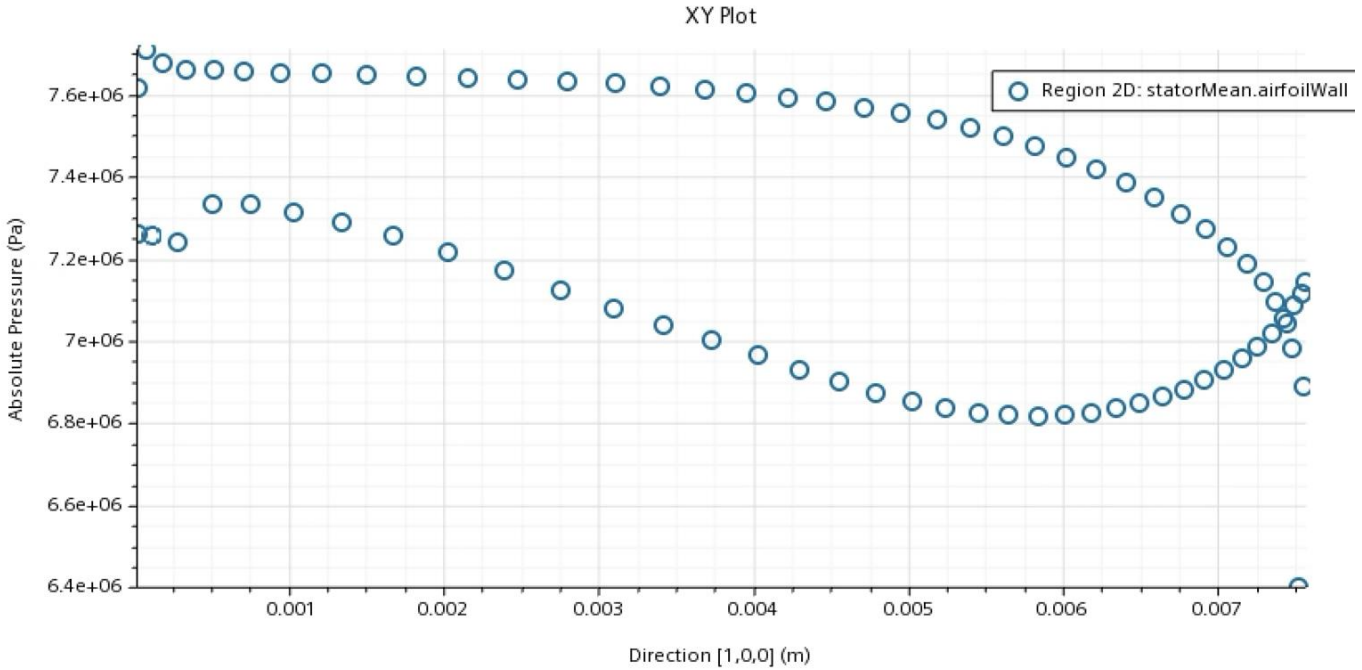


Stator Absolute Pressure and Velocity Vector Plots at Mean Radius





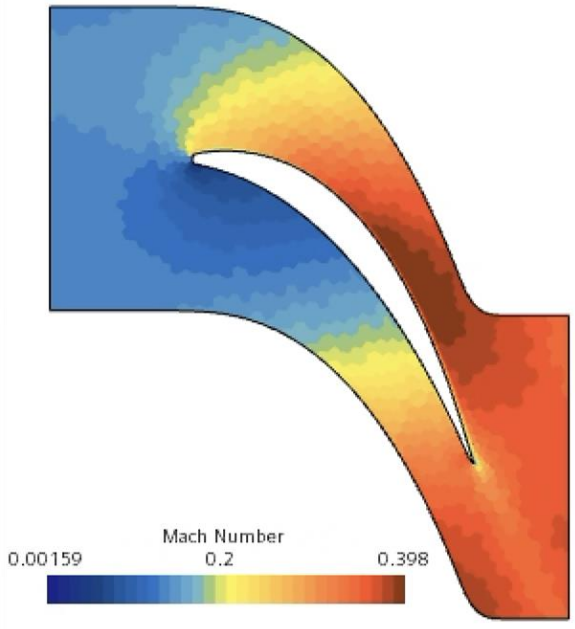
Pressure Loading - Stator (Mean Radius)

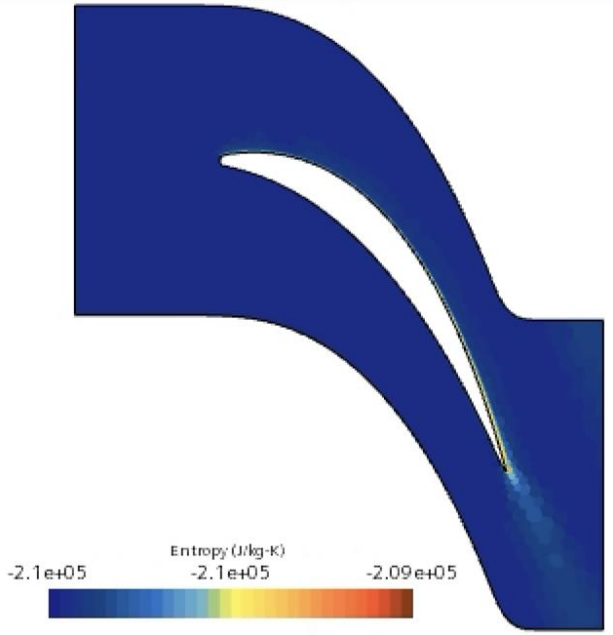


2.13.2 CFD Stator performance

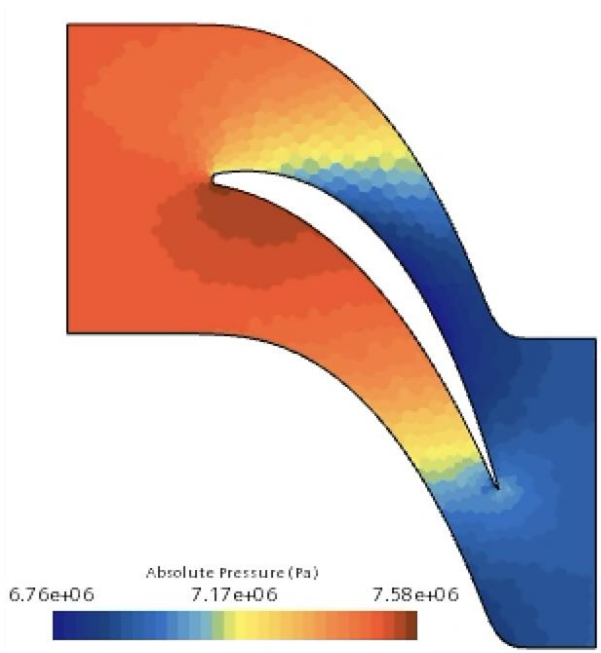
Stator - Mean Radius				
		CFD	MATLAB	Error
C2	*[m/s]*	544.4	543.6	0.15%
Cu2	*[m/s]*	476.62	490.8	2.89%
Ca2	*[m/s]*	251.4	233.7	7.57%
Discharge Angle	*[deg]*	23.43	25.46	7.97%
Mass flow (scaled for L2)	*[kg/s]*	49.3	47.6	3.64%

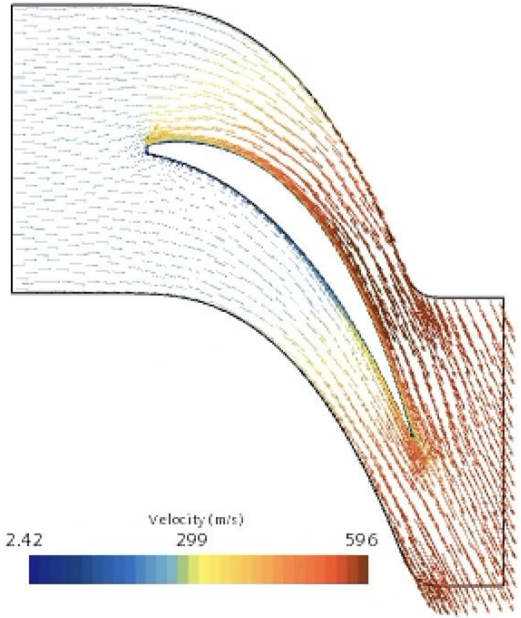
Stator Mach and Entropy Plots at Tip Radius



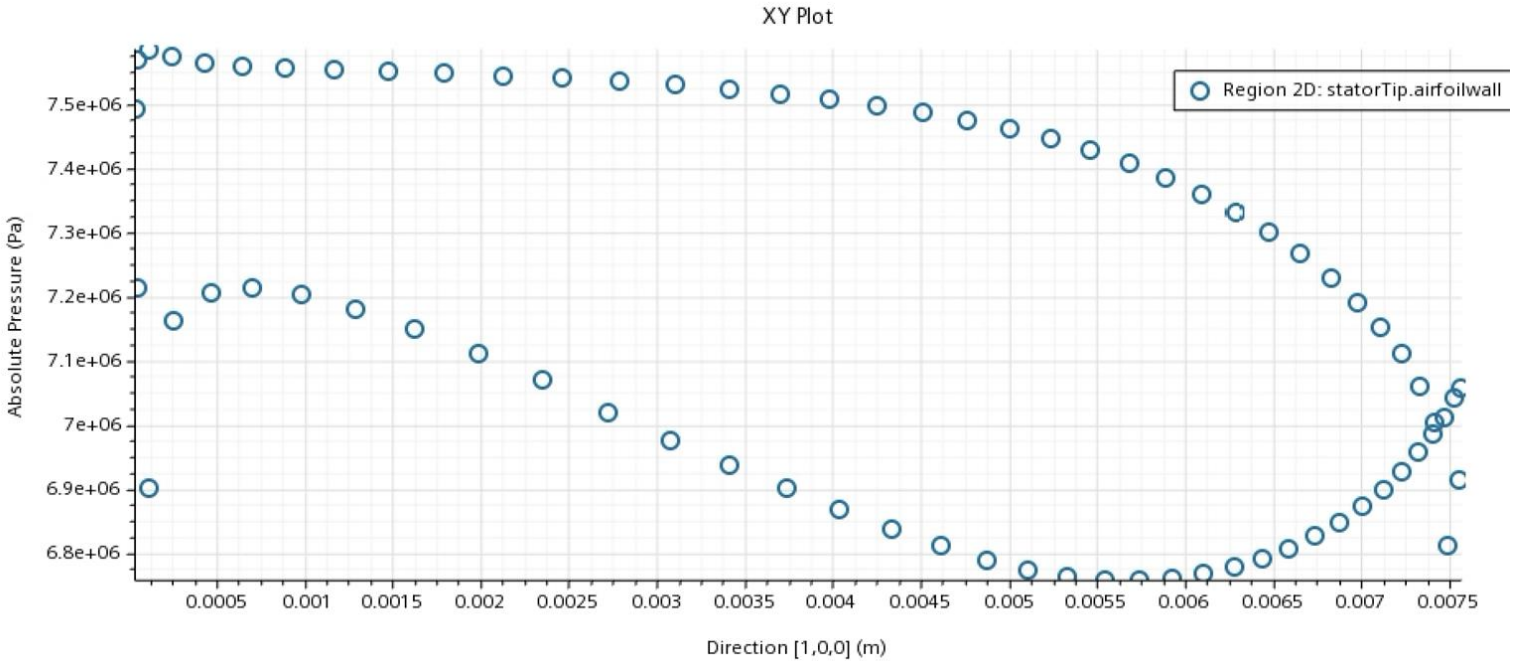


Stator Absolute Pressure and Velocity Vector Plots at Tip Radius

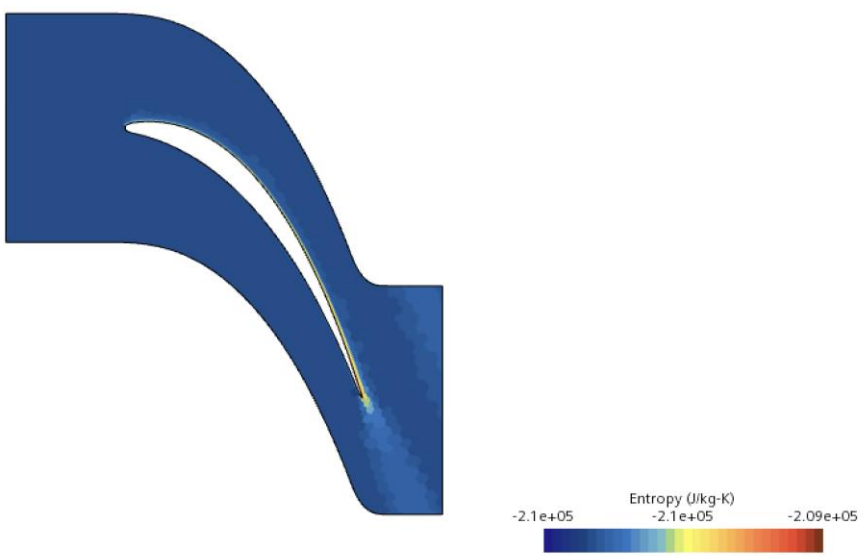


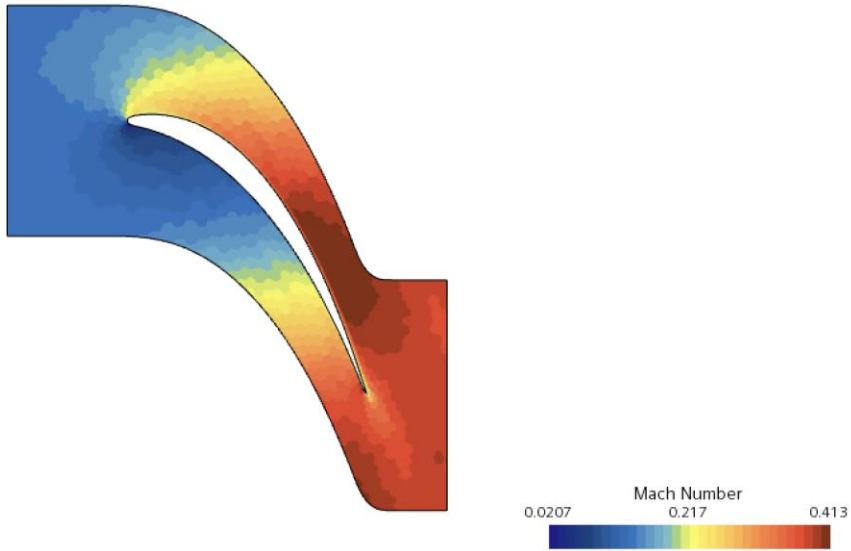


Pressure Loading - Stator (Tip Radius)

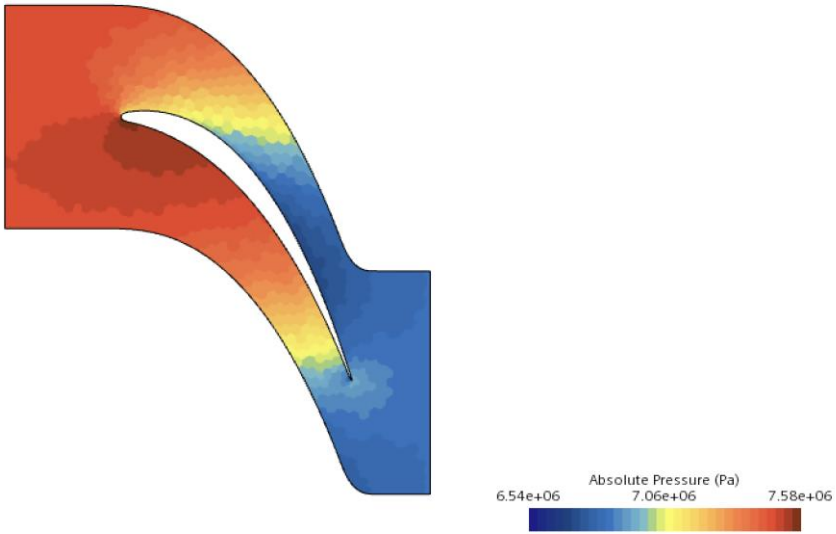


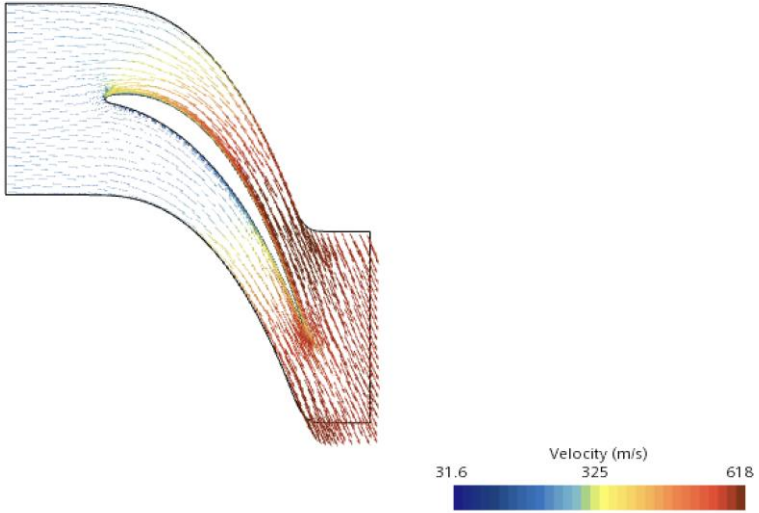
Stator Mach and Entropy Plots at Hub Radius



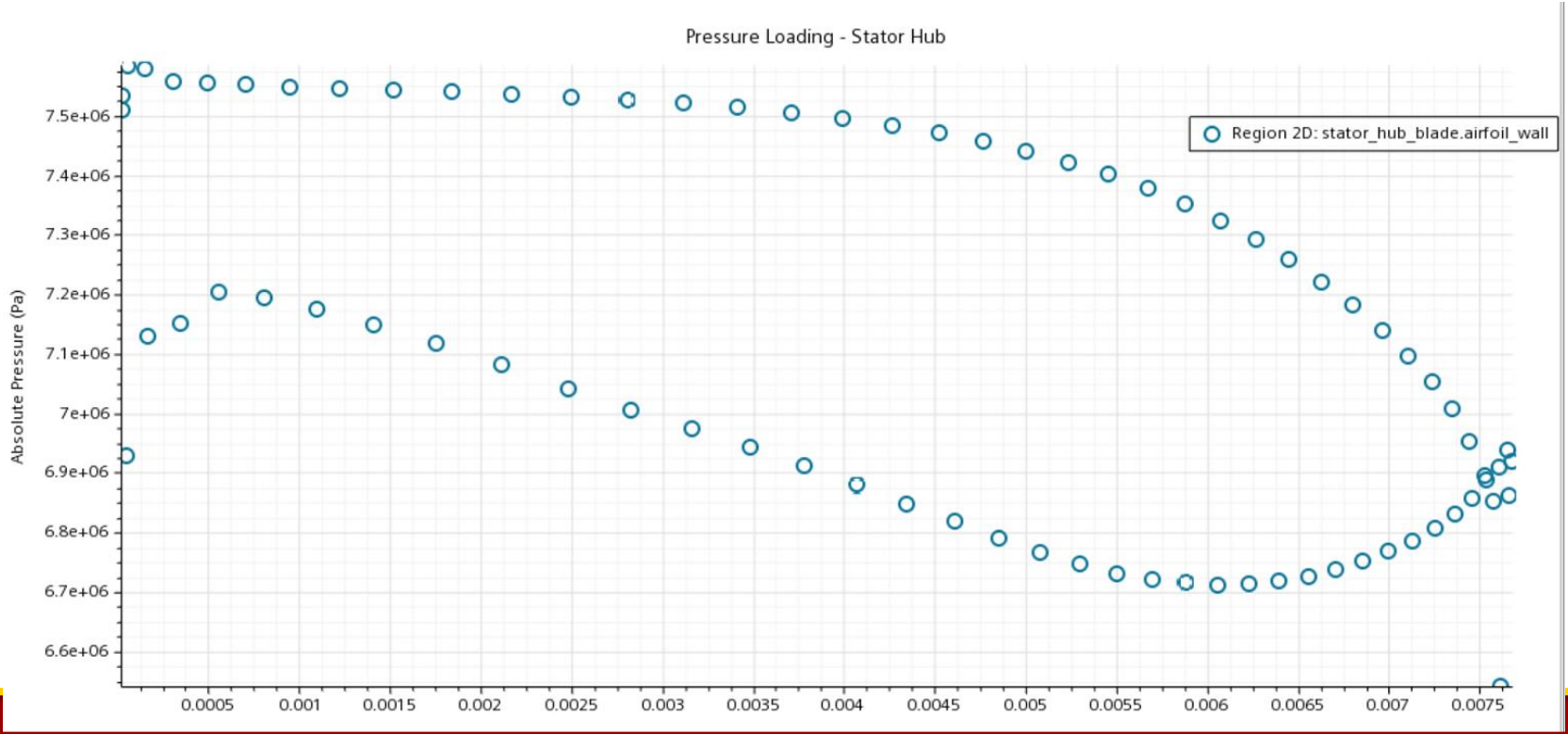


Stator Absolute Pressure and Velocity Vector Plots at Hub Radius





Pressure Loading Stator Hub



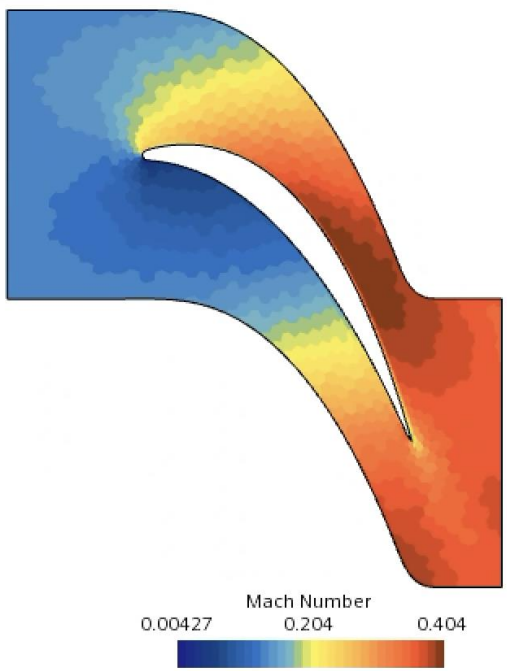
Stator Tip and Hub Values

Stator - Tip				
		CFD	MATLAB	Error
C2	*[m/s]*	531.1	523.7	1.42%
Cu2	*[m/s]*	465.9	468.6	0.58%
Ca2	*[m/s]*	241.7	233.7	3.41%
Discharge Angle	*[deg]*	24.1	26.5	9.06%
Mass flow (scaled for L2)	*[kg/s]*	49.0	47.6	2.94%

Stator - Hub				
		CFD	MATLAB	Error
C2	*[m/s]*	571.67	565.79	1.04%
Cu2	*[m/s]*	525.64	515.26	2.01%
Ca2	*[m/s]*	224.66	233.73	3.88%
Discharge Angle	*[deg]*	23.14	24.4	5.16%
Mass flow (scaled for L2)	*[kg/s]*	48.0	47.6	0.91%

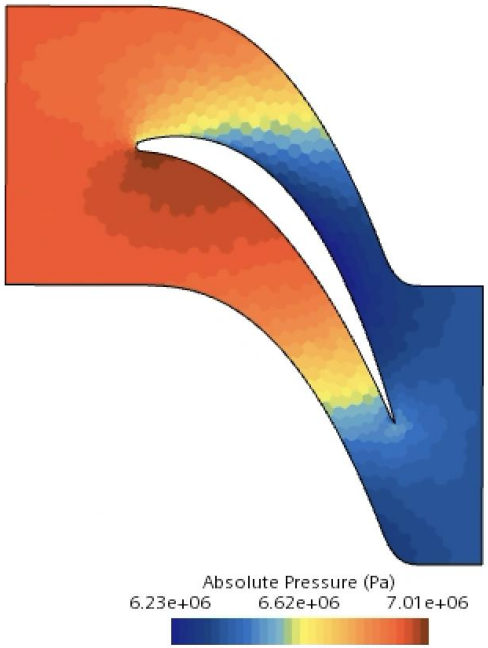
Average mass flow across all stator geometries: 48.77 kg/s

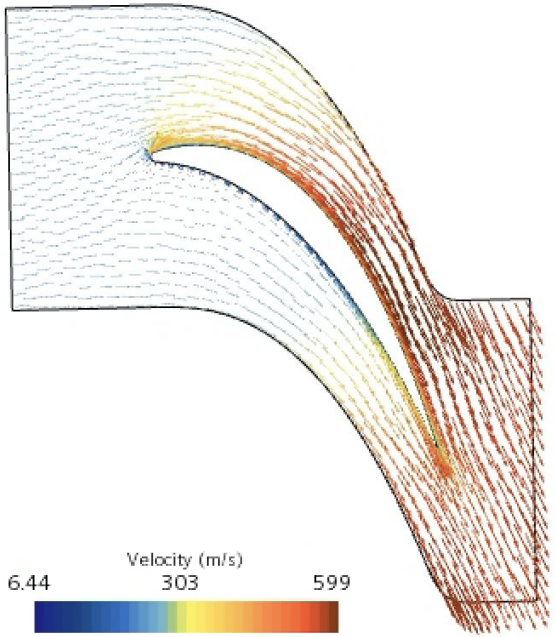
Mach and Entropy Plots at Mean Radius (Rotor)



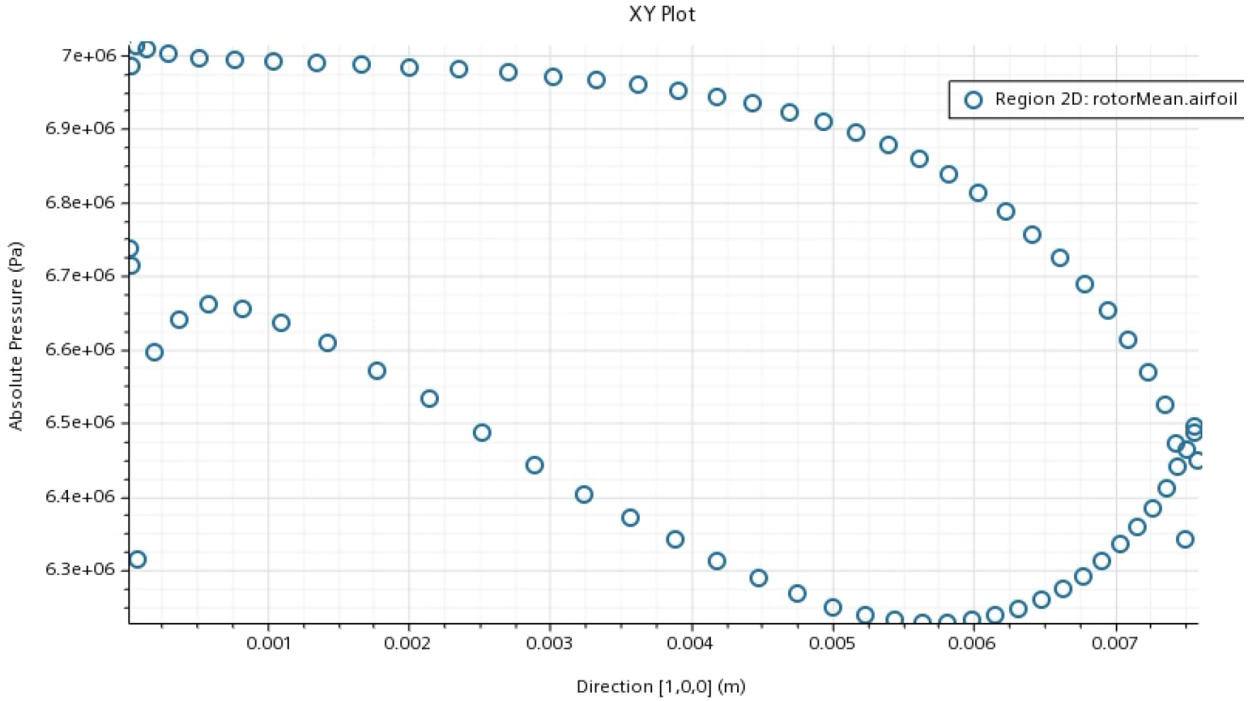


Absolute Pressure and Velocity Vector Plots at Mean Radius (Rotor)





Pressure Loading - Rotor



2.13.2 CFD Rotor performance (mean)

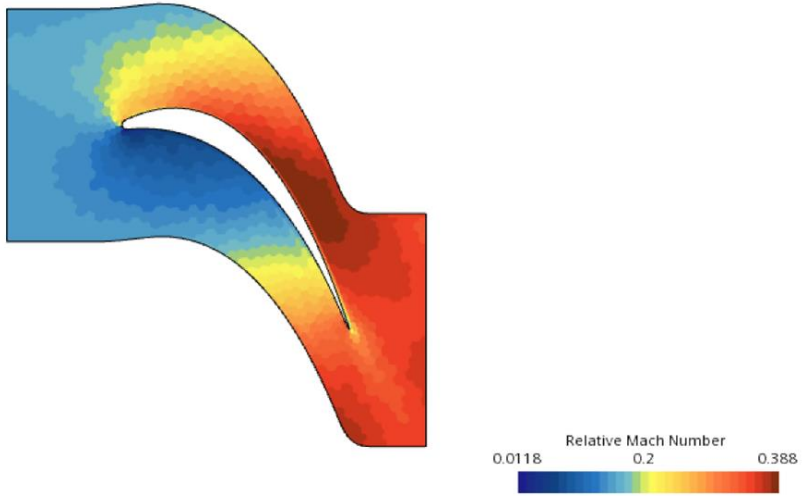
Rotor				
		CFD	MATLAB	Error
Wu2	*[m/s]*	20.5	23.4	12.4%
Wa2	*[m/s]*	176.4	233.7	24.5%
W2	*[m/s]*	204.2	234.9	13.1%
Discharge Angle	*[deg]*	23.7	26.6	11.0%
Wu3	*[m/s]*	465.9	467.5	0.3%
Wa3	*[m/s]*	252.7	233.7	8.1%
W3	*[m/s]*	536.4	522.6	2.6%
MassFlow	*[kg/s]*	48.7	47.6	2.3%

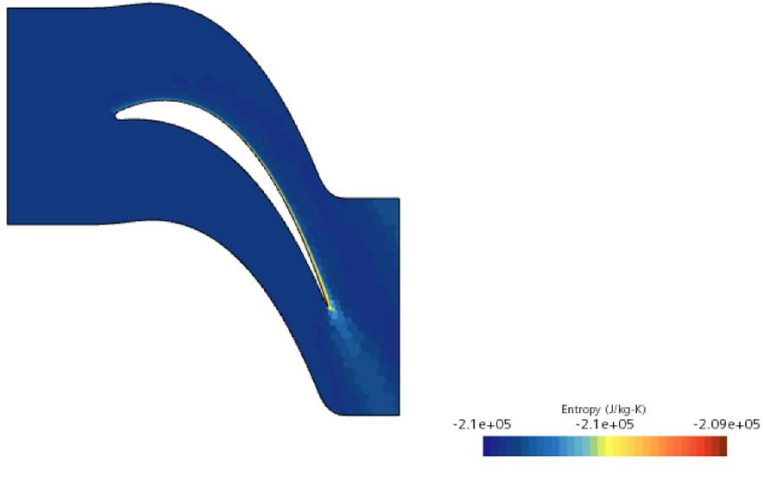
Power Comparison		
Inlet Angle	*[deg]*	84.3
Work (Euler)	*N*	227603
Torque	*N*m*	7419
Power (work and massflow)	*hp*	14870
Power (torque)	*hp*	15646
Power [MATLAB]	hp	14646

- Generally larger error values than stator values
- Wa2 with unusually high error value

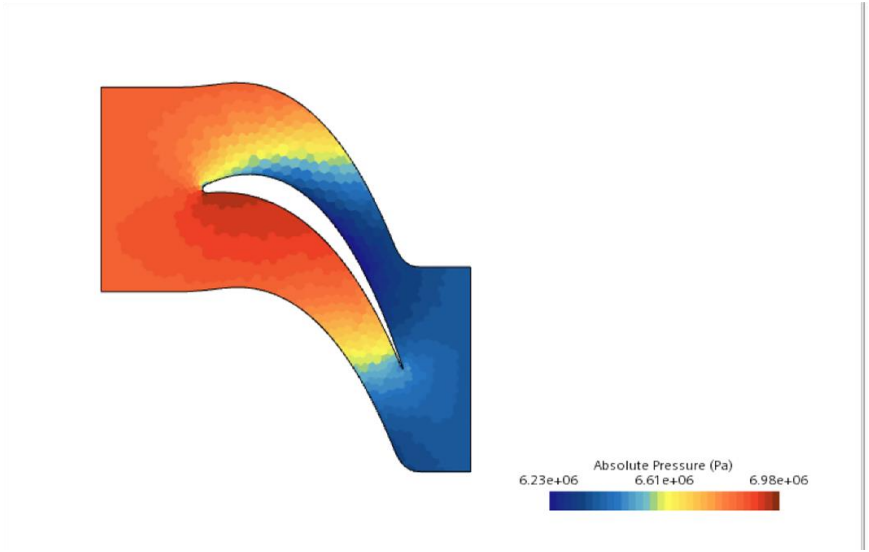
- Average of CFD power values = 14631 hp
- Very close to MATLAB value

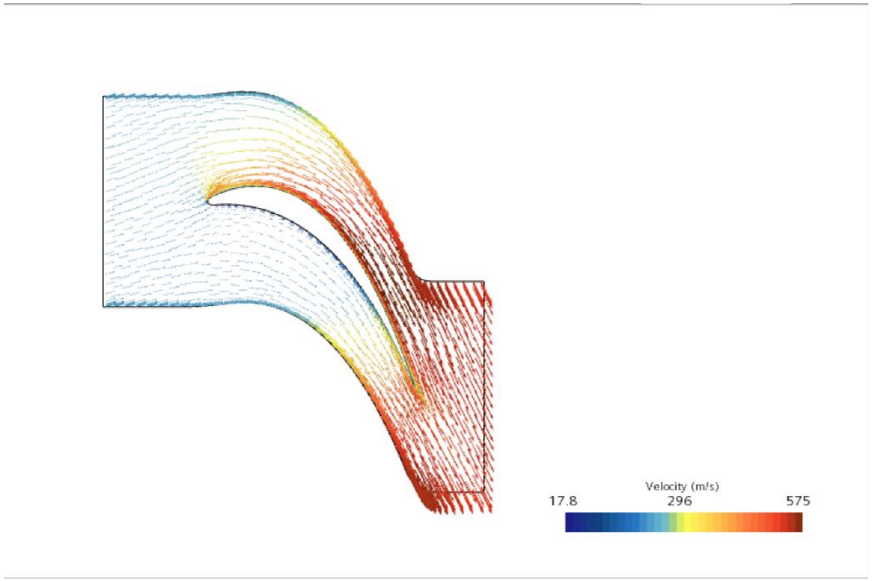
Mach and Entropy Plots at Hub Radius (Rotor)



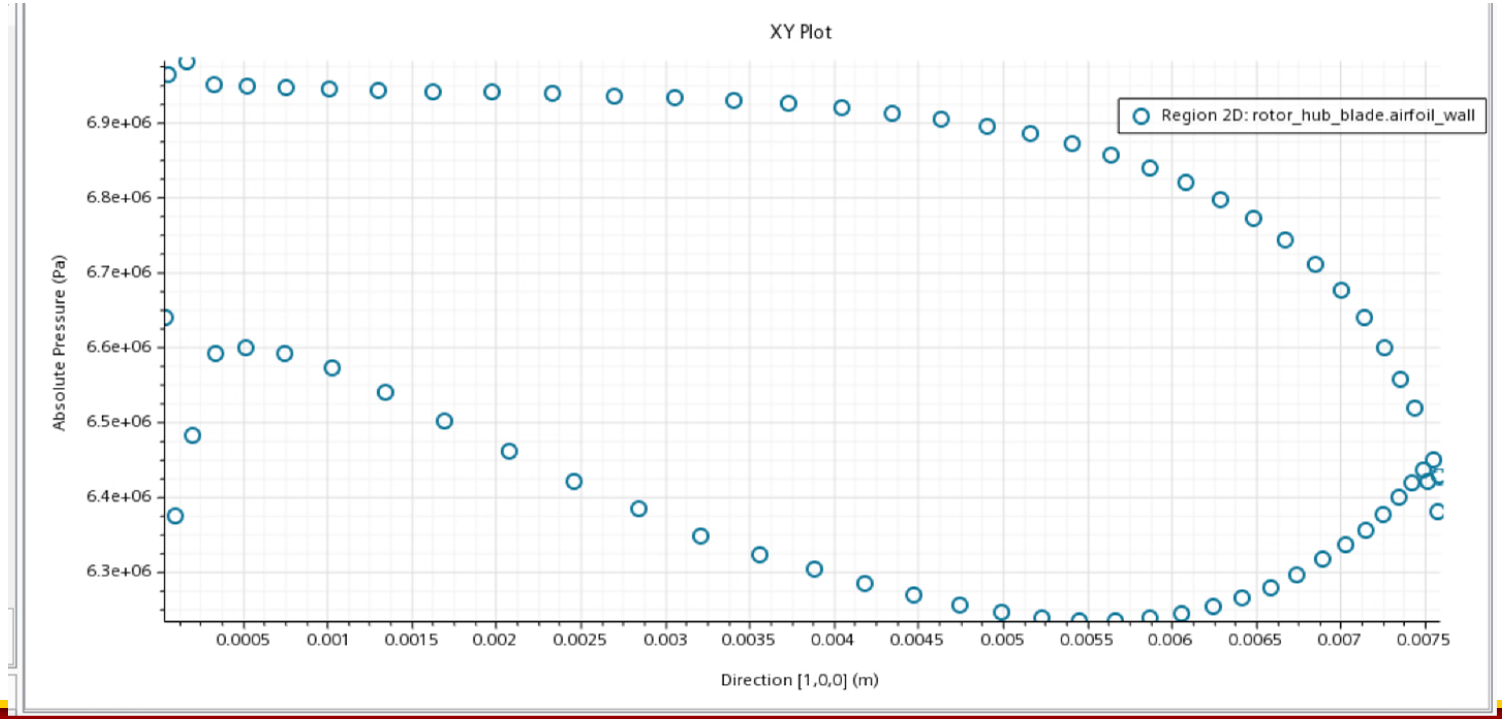


Absolute Pressure and Velocity Vector Plots at Hub Radius (Rotor)

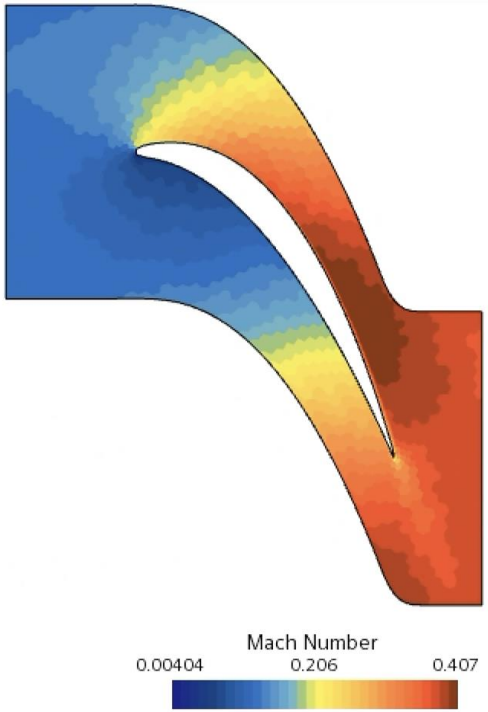




Pressure Loading Rotor Hub

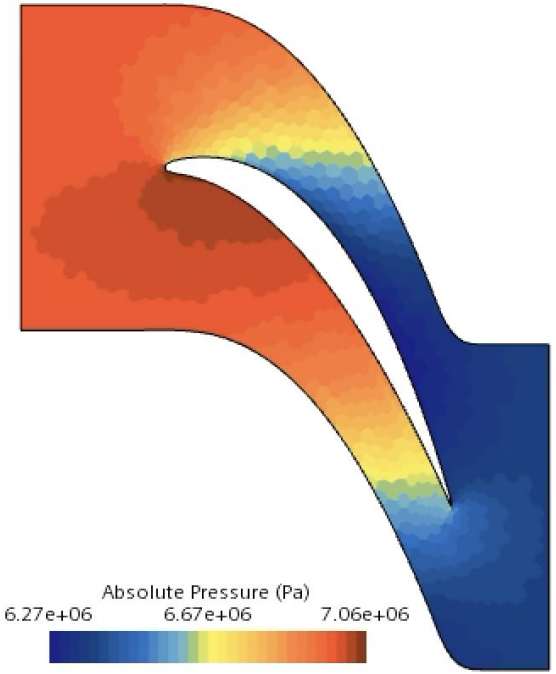


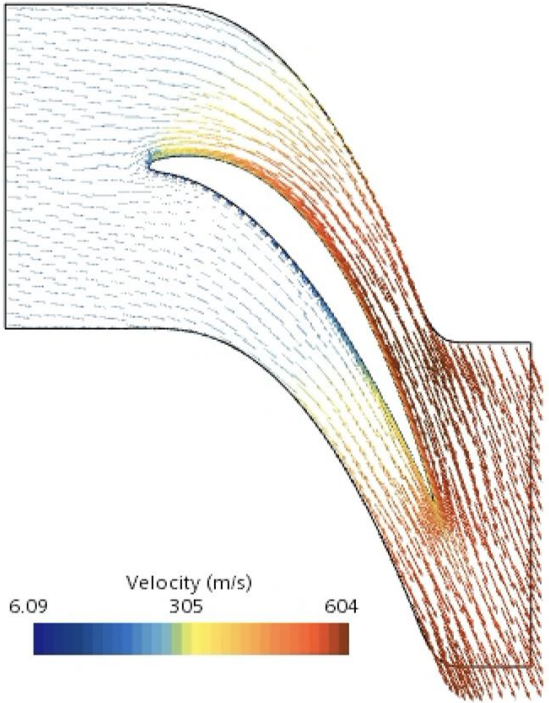
Rotor - Mach and Entropy Plots at Tip Radius



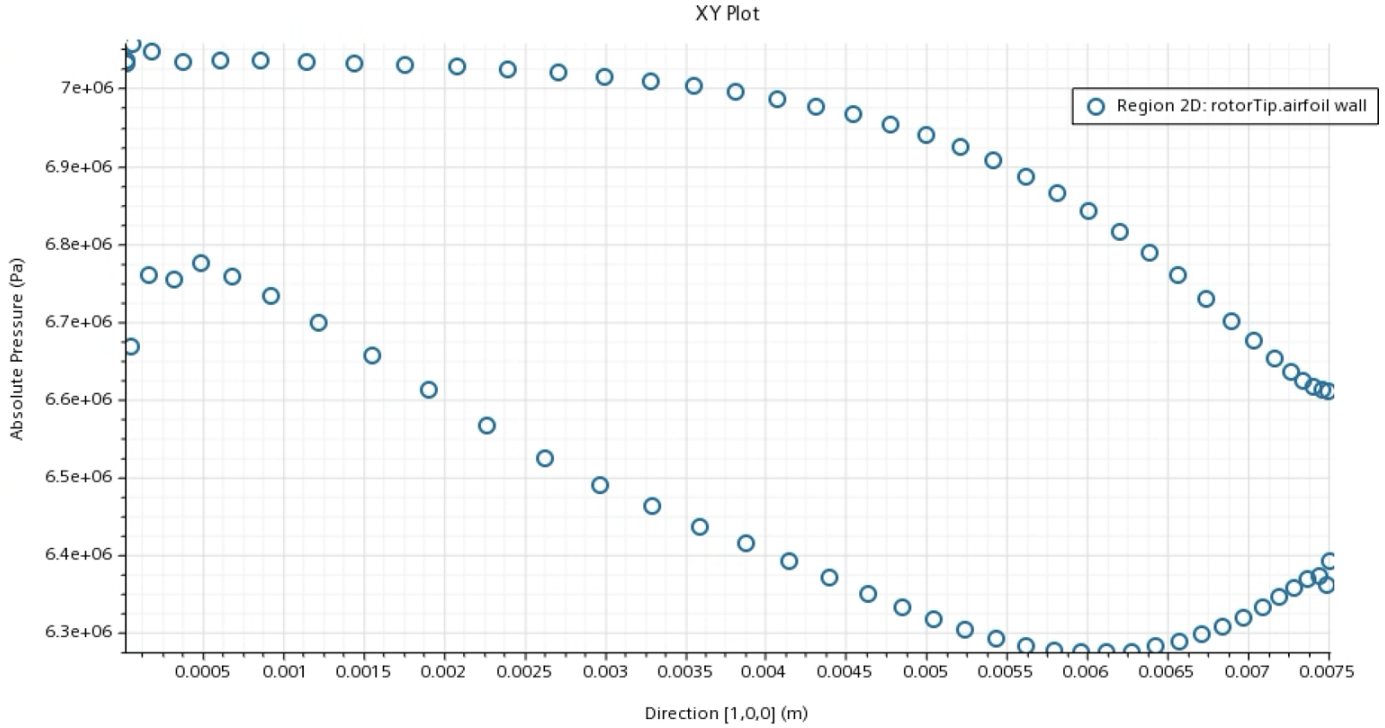


Rotor - Absolute Pressure and Velocity Vector Plots at Tip Radius





Pressure Loading - Rotor (Tip Radius)



Rotor Tip and Hub Values

Rotor - Hub				
		CFD	MATLAB	Error
Wu2	*[m/s]*	57.8	70	17.43%
Wa2	*[m/s]*	214.9	233.7	8.04%
W2	*[m/s]*	224.4	243.97	8.02%
Wu3	*[m/s]*	438.0	444	1.35%
Wa3	*[m/s]*	226.15	233.7284	3.24%
W3	*[m/s]*	517.5	501.7839	3.14%
Outlet Angle	*[deg]*	25.9	27.76	6.66%
Mass flow	*[kg/s]*	48.6	47.6	2.16%

Rotor - Tip				
		CFD	MATLAB	Error
Wu2	*[m/s]*	13.1	21.0	37.6%
Wa2	*[m/s]*	162.2	233.7	30.6%
W2	*[m/s]*	189.9	234.7	19.1%
Wu3	*[m/s]*	485.2	490.8	1.1%
Wa3	*[m/s]*	259.1	233.7	10.9%
W3	*[m/s]*	556.2	543.6	2.3%
Outlet Angle	*[deg]*	21.2	25.5	16.9%
Mass Flow	*[kg/s]*	47.8	47.6	0.5%

Power Comparison - Tip and Hub

Power Comparison - Tip		
Inlet Angle	*[deg]*	84.3
Work (Euler)	*N*	244566
Torque	*N*m*	7500
Power (work and massflow)	*hp*	15692
Power (torque)	*hp*	15815
Power [MATLAB]	hp	14646

Power Comparison - Hub		
Inlet Angle	*[deg]*	73.3
Work (Euler)	*N*	231771
Torque	*N*m*	7291
Power (work and massflow)	*hp*	15117
Power (torque)	hp	15376

Averaged Power Values

Average Power Calculations		
Massflow	[kg/s]	48.2
Work	*N*	234647
Torque	*N*m*	7403
Power (via work)	*hp*	15226
Power (via torque)	*hp*	15612
Power (MATLAB)	hp	14646